Multitarget Segmentation and Adaptive Tracking of Single-Photon LiDAR in Ultralow-SNR UAV Swarm Scene

Dongfang Guo, Jianfeng Sun, *Member, IEEE*, Xin Zhou, Sheng Chen, Yinbo Zhang, Sining Li, and Yanchen Qu

***Abstract*—Single-photon LiDAR (SPL) offers extreme detection sensitivity for long-range sensing but remains fundamentally challenged in multitarget scenarios where echoes from different objects strongly overlap under severe photon scarcity. This limitation prevents reliable perception of unmanned aerial vehicle (UAV) swarms in low-SNR environments. Here, we present a multitarget residual-enhanced adaptive tracking (MRAT) framework that resolves overlapping photon echoes by exploiting statistical differences in photon-event distributions across time-of-flight and frame domains. By modeling residual structures in photon-event histograms and enforcing spatiotemporal consistency, MRAT separates and tracks multiple UAV targets even under severe spatial overlap. Field experiments using a Geiger-mode APD (Gm-APD) SPL prototype demonstrate reliable segmentation and trajectory reconstruction of multiple UAVs at ranges up to 1 km across diverse weather conditions and heterogeneous backgrounds (sky, forest, and urban buildings). The system maintains multitarget separability at signal-to-noise ratios down to −14.0814 dB with fewer than 0.1744 average signal photons per pixel and achieves a minimum radial resolution of 4.5 m while sustaining subpixel tracking accuracy during severe target interactions and occlusions. These results provide the first experimental demonstration of SPL for UAV-swarm tracking and suggest a pathway toward long-range multitarget perception for wide-area low-altitude monitoring.**

***Index Terms*—Geiger-mode avalanche photodiode (Gm-APD), multitarget tracking, single-photon LiDAR (SPL), target segmentation, unmanned aerial vehicle (UAV) swarm detection.**

I. Introduction

UNMANNED aerial vehicle (UAV) swarms, owing to their low cost, high maneuverability, and distributed cooperative capabilities, have emerged as a critical challenge to low-altitude situational awareness and air defense systems [1], [2]. UAVs are typically small, have a low radar cross section (RCS), and are often constructed from low-reflectivity materials, resulting in extremely weak echoes that are easily submerged in strong electromagnetic clutter and interference. Moreover, the dense formation and high maneuverability of UAV swarms induce range migration, Doppler spreading, and echo superposition, collectively reducing the reliability of detection, segmentation, and continuous tracking.

Existing monitoring systems mainly rely on radar, radio frequency (RF), and electro-optical (EO) sensing, each exhibiting inherent limitations. MIMO radar performance is constrained by antenna aperture and wavelength in the angular dimensions and by signal bandwidth in range resolution [3], making it difficult to simultaneously achieve wide-field coverage and high-precision lateral resolution at kilometer-scale distances (e.g., X-band radars require approximately 36-m angular separation to resolve dual targets at 1 km) [4], [5]. Recent studies have introduced advanced signal processing and deep learning techniques to enhance radar-based target perception, particularly in polarimetric SAR imagery [6], [7]. Although these methods significantly improve classification performance under stable imaging conditions, they generally require dense signal statistics and reliable image formation, which remain difficult to guarantee for long-range, low-RCS, and highly dynamic UAV swarm scenarios. RF-based monitoring can estimate target locations through communication interception or direction-of-arrival (DoA) estimation [8], [9]. However, its detection reliability is highly sensitive to channel conditions, and localization accuracy degrades significantly under low SNR, multipath propagation, and multisource interference. Reported experiments indicate positioning errors of approximately 15 m at a range of 1.7 km, while detection performance deteriorates rapidly when the SNR falls below 0 dB [10], [11]. EO sensing provides high spatial resolution but is inherently passive and, therefore, limited to 2-D observations. Its performance is also sensitive to illumination conditions, weather, and target occlusion [12], [13]. Consequently, current sensing modalities face a fundamental tradeoff among resolution, coverage, multitarget separability, and dynamic reliability, which makes real-time perception of dense UAV swarms highly challenging.

Recently, time-correlated single-photon counting (TCSPC)-based LiDAR, combined with Geiger-mode avalanche photodiodes (Gm-APDs) or superconducting nanowire single-photon

Received 9 December 2025; revised 18 January 2026, 12 March 2026, and 15 May 2026; accepted 12 June 2026. Date of publication 17 June 2026; date of current version 29 June 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62505069, in part by the National Key Laboratory of Laser Spatial Information Foundation under Grant LSI2024WDZC04, and in part by the China Scholarship Council under Grant 202606120073. *(Corresponding author: Xin Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2026.3704221

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detectors (SNSPDs), has demonstrated remarkable sensitivity in long-range sensing, bioimaging, and non-line-of-sight imaging applications [14], [15], [16], [17]. Experimental studies have shown that single-photon LiDAR (SPL) can achieve 3-D mapping of mountainous terrain at distances exceeding 200 km with only 0.44 photons per pixel [18]. SPL arrays are capable of capturing 4-D spatiotemporal dynamic information, including range, time-of-arrival, and spatial position within the field of view (FoV) [19]. These measurements enable the estimation of size, RCS, 3-D velocity, and spatial position, thereby offering significant potential for low-altitude situational perception. Motivated by these advantages, several pioneering studies have explored the feasibility of SPL for UAV detection and tracking: Steinvall and Tulldahl [20] reported UAV profiling measurements at 1.3 km under subphotons; Guo et al. [21], [22] demonstrated real-time tracking of dynamic UAVs at 3 km using a 64×64 Gm-APD SPL. Despite these advances, SPL-based perception of UAV swarms remains largely unexplored due to a fundamental challenge arising from two coupled factors. SPL typically relies on sufficient photon accumulation over relatively long integration times to reconstruct target echoes via temporal histograms [23]. For dynamic targets, however, the integration time must be shortened to suppress motion blur, resulting in extremely sparse photon returns that are overwhelmed by background noise. In multitarget scenarios, echoes from different UAVs may overlap in the temporal domain, significantly broadening histogram distributions and rendering conventional single-peak reconstruction approaches ineffective. Consequently, reliable multitarget tracking from ultrasparse and temporally overlapped photon returns remains a major challenge for SPL-based dynamic sensing.

To overcome these limitations, we propose a multitarget residual-enhanced adaptive tracking (MRAT) framework that operates directly on sparse photon-event statistics rather than reconstructed images. By exploiting the statistical structure of photon arrivals, MRAT enables robust multitarget perception even under extremely low-SNR conditions where conventional image-centric approaches fail. The main contributions of this work are summarized as follows.

1) *Residual-Enhanced Temporal Modeling:* A novel strategy is developed to explicitly separate target-triggered photon events from background-induced statistics by exploiting global and background-triggered residuals. This mechanism significantly amplifies weak, overlapping echoes in the time domain, enabling multitarget separability with low-SNR reconstructions.
2) *Adaptive Truncation-Guided Segmentation:* An adaptive scheme is proposed, which leverages system-constrained pulse shape characteristics and spatiotemporal consistency. This effectively suppresses nonphysical prepeaks and background artifacts, resolving strongly superposed noise photons under extreme photon sparsity.
3) *Scale-Adaptive Event-Driven Tracking:* MRAT integrates segmentation with a scale-adaptive tracking mechanism that combines optical-flow estimation, BANSAC-guided template matching, and pyramid-based search, enabling stable 3-D tracking under depth-induced scale variation and intermittent photon returns.

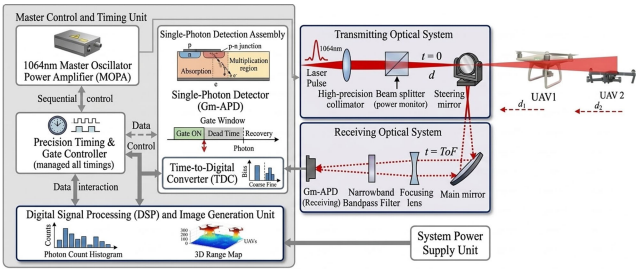


Fig. 1. Detection schematic of Gm-APD SPL.

The proposed MRAT framework is validated through extensive field experiments using a 64×64 Gm-APD SPL prototype. Up to three UAVs are simultaneously tracked to emulate swarm scenarios at ranges of 400 m under strong illumination (90 530 lux). MRAT achieves stable multitarget segmentation at SNRs as low as -14.0814 dB, corresponding to an average of 0.1744 photons per pixel. The SPL achieves a minimum resolvable radial separation of 4.5 m, maintaining subpixel tracking accuracy during severe target interactions and occlusions, and reliable target discrimination even under complete lateral overlap. With a 2000-ns gating window, the effective temporal span theoretically supports the separation of up to 61 UAVs along the FoV. Further experiments conducted in rainy and foggy environments across forested, urban-building, and open-sky scenes demonstrate the robustness of MRAT. In all cases, reliable multitarget segmentation is maintained despite significant signal photon and SNR attenuation. Comparative evaluations against representative tracking approaches, including mean-shift Kalman fusion, particle filtering, joint probabilistic data association, multiple hypothesis tracking, and probability hypothesis density filtering, demonstrate that the proposed MRAT framework consistently achieves lower mean and RMSE tracking errors. Even under challenging conditions involving target projection overlap and trajectory occlusion, the proposed method maintains a mean tracking error below 0.92 pixels at an SNR of -14.0814 dB, corresponding to improvements of more than 43% and 26% compared with mean-Kalm and particle filtering, respectively. These results demonstrate that multitarget perception can be achieved directly from ultrasparse and temporally overlapped photon returns, thereby establishing a photon-statistics-driven sensing paradigm for dynamic SPL perception.

The remainder of this article is organized as follows. Section II introduces the dynamic Gm-APD detection model for multitarget sensing. Section III presents the proposed MRAT segmentation and tracking framework. Section IV presents the experimental results and performance analysis. Finally, Section V concludes and discusses future research directions.

II. Gm-APD Multitarget Dynamic Detection Model

The detection process of the Gm-APD SPL is based on high-frequency photon detection events and its statistical modeling, as illustrated in Fig. 1.

Multiple-UAV targets illuminated by the laser beam can be modeled as point targets [15]. For the kth target, the received echo power can be expressed as [14]

$$P_{r,k}(t) = \frac{4P_t(t-\tau_k)}{\pi\theta_B^2 R_k^2}\rho_k A_{T,k}\frac{A_r}{\pi R_k^2}\eta_{atm}\eta_{sys} \quad (1)$$

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where P_t is the transmitted laser power temporal profile; ρ_k is the reflectivity of the kth target; θ_B is the laser divergence angle; R_k is the target range; $A_{T,k}$ is the effective area of the target; A_r is the effective area of the receiving lens; $\tau_k = 2R_k/c$ is the round-trip time delay for the kth target, with c being the speed of light; η_{atm} is the two-way atmospheric transmittance; and η_{sys} is the system efficiency, including optical transmittance and detector response.

Let the arrival time of echo photons be denoted by the laser pulsewidth t_d with a time resolution Δt, corresponding to the time-to-digital converter (TDC) bin width. Assuming uniform laser energy distribution and negligible effects from speckle and turbulence, the mean photon arrival number $\mu_i^{(k)}$ within the ith time bin is

$$\mu_i^{(k)} = \frac{\eta_q}{h\nu} \int_{i\Delta t}^{(i+1)\Delta t} P_{r,k}(t)\, dt \tag{2}$$

where η_q is the quantum efficiency of the Gm-APD and $h\nu$ is the single-photon energy.

Background noise, primarily originating from ambient solar radiation and the internal dark counts of the detector, can be modeled as a continuous constant arrival rate β

$$\beta = \frac{\eta_q P_{bg}}{h\nu} + f_{DC} \tag{3}$$

where f_{DC} is the dark count rate (DCR) of the Gm-APD and P_{bg} is the received background optical power, which can be calculated by

$$P_{bg} = I_{bg} A_r \Omega_R \Delta\lambda \eta_r \tag{4}$$

where I_{bg} is the background solar spectral radiance, Ω_R is the receiver FoV solid angle, $\Delta\lambda$ is the spectral bandwidth of the optical narrowband filter, and η_r is the optical efficiency of the receiver path.

To describe the temporal distribution of echo photons, a system-equivalent impulse response $s_k^{\text{eff}}(t)$ is introduced, incorporating the effects of laser pulsewidth, target scattering broadening, and system timing jitter, normalized as $\int_{-\infty}^{\infty} s_k^{\text{eff}}(t)dt = 1$. Accordingly, the mean photon arrival rate $\lambda_k(t)$ for the kth target in the time domain is

$$\lambda_k(t) = \int_{i\Delta t}^{(i+1)\Delta t} \mu_i^{(k)} s_k^{\text{eff}}(t - \tau_k)\, dt. \tag{5}$$

The total photon arrival rate in the scene is

$$\begin{aligned} \Lambda(t) &= \sum_k \lambda_k(t) + \beta\Delta t \\ &= \int_{i\Delta t}^{(i+1)\Delta t} \left(\sum_k \mu_i^{(k)} s_k^{\text{eff}}(t - \tau_k) + \beta \right) dt. \end{aligned} \tag{6}$$

Consequently, the mean background photon number within the ith time bin can be expressed as $\mu_i^{(bg)} = \beta \cdot \Delta t$.

For scenarios with multiple targets, the cumulative photon arrival rate μ_i is given by

$$\mu_i = \int_{i\Delta t}^{(i+1)\Delta t} \Lambda(t)\, dt = \sum_k \mu_i^{(k)} + \mu_i^{(bg)}. \tag{7}$$

The Gm-APD operates in a trigger-detection mode, where the arrival of one or more photons initiates avalanche multiplication, generating a saturated electrical pulse and, thereby, registering a photon-trigger event [19]. The single-pulse triggering probability $p_i(h)$ for the ith time bin is given by

$$p_i = 1 - e^{-\mu_i} \tag{8}$$

which indicates that Gm-APD detection follows a Bernoulli trial [24]. After repeating the pulse for N times, the number of triggers in the ith bin, K_i, follows a binomial distribution

$$\begin{cases} K_i \sim \text{Binomial}(N, p_i) \\ \mathbb{E}[K_i] = N p_i \\ \text{Var}[K_i] = N p_i (1 - p_i). \end{cases} \tag{9}$$

Under low-photon conditions ($\mu_i^{(k)} \leq 1$), the single-pulse triggering probability can be approximated linearly as $p_i \approx \mu_i$, and the expected number of triggers is

$$\mathbb{E}[K_i] = N p_i \approx N \mu_i. \tag{10}$$

Under the low-photon approximation, the global photon-event histogram provides an approximate representation of target intensity and range. The histogram height is proportional to the mean photon number $\mu_i^{(k)}$ in the corresponding time bin, while the peak position indicates the target round-trip delay τ_k. Multiple targets produce multiple peaks with amplitudes proportional to their photon returns. Owing to the finite width of the system-equivalent impulse response, individual peaks exhibit temporal broadening; targets can be resolved when their separations exceed the impulse-response width. Although Gm-APD saturation may introduce nonlinear peak amplitudes at high photon arrival rates, the histogram still preserves the statistical structure of multitarget returns, providing the basis for multitarget detection.

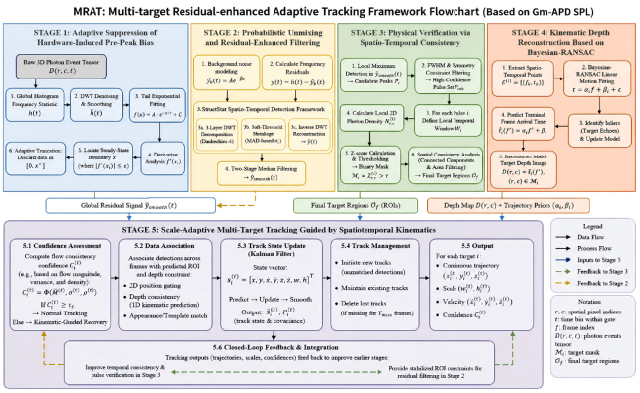


Fig. 2. Overview of the proposed MRAT framework.

III. MRAT Framework Design

Under UAV swarm detection and tracking scenarios, the performance of SPL is fundamentally governed by photon statistics rather than conventional image contrast. As discussed in Section II, the observed photon arrival rate μ_i is a superposition of multitarget echoes and background noise. Since the number of detected triggers $\mathbb{E}[K_i]$ follows a Binomial distribution with triggering probability p_i, multitarget extraction is inherently a statistical unmixing problem.

We propose the MRAT framework, as illustrated in Fig. 2. The framework is built upon the photon triggering probability model and consists of five sequential stages: (I) adaptive

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suppression to restore the statistical baseline; (II) residual-enhanced filtering for multitarget extraction; (III) spatiotemporal consistency verification for false-alarm suppression; (IV) Bayesian random sample consensus (BANSAC)-based kinematic reconstruction to eliminate motion blur and fitting the instantaneous depth positions of individual UAV target; and (V) scale-adaptive dynamic tracking performed independently on each UAV target segmented by MRAT. Together, these stages establish an integrated processing pipeline from stochastic photon-trigger events to reliable multitarget segmentation and subsequent dynamic tracking.

A. Adaptive Suppression of Hardware-Induced Prepeak Bias

According to the Gm-APD detection model, the single-pulse triggering probability p_i should ideally reflect the instantaneous photon flux. However, during the initial gate-opening of the SPL, hardware-level imperfections—such as avalanche overshoot, gate delay, and APD memory effects [23], [25]—induce nonphysical electrical pulses. These artifacts force p_i into a near-saturated triggering regime, producing an abnormally high prepeak in the global frequency distribution histogram. If left unsuppressed, this massive noise floor would completely overwhelm the genuine, extremely weak target-triggered pulses during subsequent peak detection, severely compressing the effective imaging dynamic range.

To correct this systematic probability deviation, we first design an adaptive truncation strategy based on global photon-frequency modeling. Let the discrete photon events be represented by a 3-D tensor $D(r, c, t)$, where (r, c) denotes the pixel location and t the photon arrival time within the gating window T_{gate}. Aggregating all photon arrivals along the temporal axis yields a global event set

$$T = \{t_i\}_{i=1}^{N}, \quad t_i \in \left[0, T_{\text{gate}}\right]. \tag{11}$$

A temporal histogram is constructed by accumulating the triggering photon-event frequencies of all pixels over the entire temporal domain

$$h(t_k) = \sum_{i=1}^{N} \delta(t_i - t_k), \quad k = 1, 2, \ldots, L \tag{12}$$

which is subsequently smoothed using wavelet-based denoising to suppress stochastic fluctuations.

The decay region following the dominant hardware-induced peak is modeled using an exponential function

$$f(x) = A_0 e^{-\lambda x} + C \tag{13}$$

where A_0, λ, and C, respectively, denote the initial peak amplitude, transient decay rate, and steady-state noise level, all of which are dynamically estimated from the triggering photon-event statistics via linear least-squares regression in the logarithmic domain.

Physically, the hardware-induced prepeak manifests as a high-gradient transient pulse, whereas the subsequent steady-state noise floor exhibits gentle, Poisson-governed variations. The gate-on transient artifact is considered fully dissipated when its residual amplitude ($A_0 e^{-\lambda x}$) submerges beneath the intrinsic fluctuation of the steady-state Poisson background noise, which is given by $\sigma_{bg} = \sqrt{C}$. By enforcing this physically motivated boundary condition ($A_0 e^{-\lambda x} = \sqrt{C}$) and substituting it into the analytical derivative of the decay model ($|f'(x)| = -A_0 \lambda e^{-\lambda x}$), a fully adaptive gradient threshold ε_0 is rigorously derived as

$$\varepsilon_0 = \lambda \sqrt{C}. \tag{14}$$

Both λ and C are dynamically updated per frame from the histogram fitting; this parameter-free mathematical threshold self-adjusts to varying ambient illuminations. The exact "steep-to-flat" geometric transition boundary x^*, which distinguishes the transient hardware artifact from the stable noise floor, is identified as the first instance where the gradient falls below this adaptive threshold

$$x^* = \min\left\{x_i \,\middle|\, \left|f'(x_i)\right| \le \varepsilon_0\right\}. \tag{15}$$

All temporal channels within $[0, x^*]$ are subsequently truncated to eliminate the dominating prepeak noise, preventing the target-triggered pulses from being overwhelmed. Consequently, this adaptive truncation strategy restores the effective temporal dynamic range and establishes a reliable, physically constrained statistical baseline for the subsequent multitarget echo separation.

B. Probabilistic Unmixing and Residual-Enhanced Filtering

With the baseline probability restored, the subsequent challenge is extracting the target components $\sum \mu_i^{(k)}$ from the cumulative rate μ_i. Because target arrivals follow localized temporal intensity variations, while background events $\mu_i^{(bg)}$ exhibit stable statistical behavior [26], [27], we explicitly isolate their probabilistic characteristics.

After adaptive gating, the global histogram $\tilde{h}(t)$ reflects the underlying physical trend encoded by $\Lambda(t)$. In the steady-state region devoid of targets, the background triggering probability approximately follows an exponential decay

$$y_{bg}(t) = A_1 e^{-\beta t} \tag{16}$$

which can be equivalently expressed in the logarithmic domain as

$$\log y_{bg}(t) = \log A_1 - \beta t + \varepsilon(t) \tag{17}$$

where A_1 represents the initial background rate after gate opening, β characterizes the decay coefficient, and $\varepsilon(t)$ models stochastic fluctuations. By utilizing this logarithmic representation, the background parameters (A_1, β) can be rapidly and deterministically estimated via linear least-squares regression, avoiding the high computational overhead and local-minima issues associated with nonlinear fitting.

According to the additive property of the photon arrival rate, the residual signal intrinsically represents the sum of the multitarget mean photon numbers

$$y(t) = \tilde{h}(t) - y_{bg}(t). \tag{18}$$

To mitigate the remaining pseudopulses arising from statistical variance $\text{Var}[K_i]$, a multiscale residual enhancement strategy using discrete wavelet transform and adaptive soft-thresholding is applied. This extracts structurally consistent

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target probabilities from the ambient noise floor. This process suppresses nonstructural noise while preserving the temporal symmetry and amplitude characteristics of true target echoes, producing a structurally consistent residual signal for subsequent multitarget segmentation and reconstruction.

C. Physical Verification via Spatiotemporal Consistency

Although the residual signal $y(t)$ isolates potential target time-of-flight (ToF), statistical anomalies can still manifest as residual peaks. Therefore, these temporal candidates must be physically verified against the system-equivalent impulse response $s_k^{\text{eff}}(t)$ defined in (5).

Considering a Gaussian laser pulse and detector timing jitter $J(t) = 1/((2\pi)^{1/2}\sigma_j)\exp(-t^2/(2\sigma_j^2))$, the resulting SPL impulse response limits the temporal concentration of any valid echo

$$s_{\text{sys}}(t) = \frac{1}{\sqrt{2\pi}\sigma_{\text{sys}}}\exp\left[-\frac{(t-\tau_0)^2}{2\sigma_{\text{sys}}^2}\right] \tag{19}$$

where τ_0 denotes the round-trip delay and $\sigma_{\text{sys}}^2 = \sigma_L^2 + \sigma_j^2$.

The genuine target produces a symmetric temporal peak in the residual signal, whose width is determined by the system impulse response, $\text{FWHM}_{\text{tgt}} = 2(2\ln 2)^{1/2}\sigma_{\text{sys}}$. In contrast, residual noise follows an asymmetric distribution $p_{\text{noise}}(t) = \beta + \alpha e^{-\alpha t}$. This distinction enables reliable extraction of candidate pulses $\mathcal{P}_c = \{\mathcal{P}_i = (t_i, h_i, w_i)\}_{i=1}^{M_{\text{peak}}}$.

For each validated pulse, photon events are accumulated within a temporal window $\mathcal{W}_i = [t_i - w_i/2, t_i + w_i/2]$, where w_i is directly determined by FWHM_{tgt} rather than treated as a tunable parameter. This physically constrained window captures target-induced temporal broadening while suppressing background noise. The resulting spatial aggregation is evaluated using the standardized deviation

$$Z_{r,c}^{(i)} = \frac{N_{r,c}^{(i)} - \mu_i}{\sigma_i} \tag{20}$$

forming a binary mask $\mathcal{M}_i = \{(r,c) \mid Z_{r,c}^{(i)} > \Gamma_{\text{th}}^{(i)}\}$, where the adaptive physical threshold $\Gamma_{\text{th}}^{(i)}$ is rigorously established guided by the constant false alarm rate (CFAR) criterion.

Unlike traditional microwave radar, where thermal noise dictates Gaussian or Rayleigh statistics, the ambient background events in SPL are fundamentally governed by photon shot noise and detector dark counts, strictly adhering to spatial Poisson statistics. Due to localized background inhomogeneities and detector array nonuniformity, the spatial noise floor inherently exhibits overdispersion [28], [29]. To robustly capture this practical distribution, the dynamic background threshold is formulated as

$$\Gamma_{\text{th}}^{(i)} = \mu_{\text{bg}}^{(i)} + k_{\text{std}}\sigma_{\text{bg}}^{(i)} \tag{21}$$

where $\mu_{\text{bg}}^{(i)}$ and $\sigma_{\text{bg}}^{(i)}$ are the spatial mean and sample standard deviation of the photon counts across the entire array within the ith temporal window, respectively.

The multiplier k_{std} is strictly constrained by both the focal plane array scale and the fundamental statistical properties of the ultralow-SNR regime. For microwave radar governed by continuous Gaussian thermal noise, a 3σ threshold yields a false alarm rate of $P_{\text{fa}} \approx 1.35 \times 10^{-3}$ [30]. However, applying this statistical bound directly to a 64×64 SPL array (4096 pixels) would produce approximately 5.5 false alarm pixels per frame, severely corrupting subsequent photon-event association triggered by multitarget.

More crucially, unlike Gaussian noise, the sparse background photon counts in our regime strictly follow a discrete Poisson distribution. In low-mean scenarios, the Poisson distribution exhibits a heavily skewed right tail. Consequently, the actual probability mass extending beyond a standard 3σ bound is significantly larger than Gaussian approximations predict. To rigorously suppress spatial false alarms, we elevate the statistical confidence boundary to $k_{\text{std}} = 5$. Due to the discrete and skewed nature of the Poisson tail, this 5σ boundary empirically bounds the actual false alarm rate to the order of $10^{-4} \sim 10^{-5}$ depending on the local background mean. This dynamically restricts the expected number of spatial false alarms per frame to less than 1 (e.g., $\ll 0.2$ for $\mu_{\text{bg}} = 2$), completely isolating genuine highly concentrated target clusters from the sparse Poisson noise floor without requiring manual empirical tuning.

By extracting $\mu_{\text{bg}}^{(i)}$ and $\sigma_{\text{bg}}^{(i)}$ dynamically from the raw photon-event tensor, this analytical bounding provides an environment-adaptive mechanism. It effectively guarantees a probabilistically negligible spatial false alarm rate, completely isolating highly concentrated target photon clusters from the sparse Poisson noise floor without requiring manual parameter tuning across varying ambient illuminations.

Crucially, genuine target returns are not uniformly distributed but are spatially governed by the optical point-spread function (PSF) of the system's architecture

$$S_{\text{tgt}}(x,y) = A\exp\left[-\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma_{\text{PSF}}^2}\right]. \tag{22}$$

Spatial connectivity analysis is applied to $\mathcal{M}_i$ to enforce this physical distribution, yielding the final set of physically candidate target regions $\mathcal{O}_f = \{\mathcal{O}_j\}_{j=1}^{\mathcal{M}_i}$.

D. Kinematic Depth Reconstruction Based on BANSAC

The preceding stages successfully unmix target-associated photon events from the noise, yielding highly confident spatial target regions. However, an inherent contradiction fundamentally limits dynamic detection under extreme photon scarcity: ensuring sufficient SNR requires accumulating photon events across multiple frames, yet the high-speed radial motion of the UAV causes the photon ToF to migrate continuously across the temporal bins over successive frames. Simply aggregating these events directly would result in severe range migration and motion blur, making it impossible to determine the target's precise instantaneous depth.

To resolve this spatiotemporal contradiction, the discrete photon events $\mathcal{E}^{(j)} = \{(f_k, t_k)\}_{k=1}^{N_j}$ aggregated within the jth extracted spatial target region $\mathcal{O}_j$ must be kinematically constrained. The BANSAC approach is introduced to robustly fit a linear spatiotemporal motion model

$$t = \alpha_j f + \beta_j + \varepsilon. \tag{23}$$

In each iteration, BANSAC randomly samples minimal data pairs from $\mathcal{E}^{(j)}$ to generate a trajectory hypothesis. Crucially,

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the evaluation of the consensus set (inliers) is not based on an empirical parameter but is strictly constrained by the system's physical impulse response. Specifically, a photon event (f_k, t_k) is classified as a valid kinematic inlier if and only if its temporal residual against the hypothesized trajectory satisfies

$$\left|t_k - \left(\alpha_j f_k + \beta_j\right)\right| \leq \frac{w_j}{2} \tag{24}$$

where w_j is the extracted temporal width of the jth target peak. This physical boundary ensures that only photon events falling strictly within the target's temporal broadening envelope are considered valid trajectory contributors. By maximizing the inlier consensus set across iterations, the optimal kinematic parameters (α_j, β_j) are extracted, explicitly representing the target's interframe radial motion rate and baseline temporal delay while rejecting residual background events. Evaluating the fit kinematic model at the current target frame f' yields the instantaneous relative photon arrival time of the target, given by $\hat{t}_j(f') = \alpha_j f' + \beta_j$.

It is crucial to note that the Gm-APD in SPL operates in a range-gated mode to suppress atmospheric backscatter and prevent premature detector saturation [31]. Consequently, the temporal value $\hat{t}_j(f')$ recorded by the TDC represents the relative delay from the gate opening rather than the absolute photon ToF. Taking the preconfigured hardware gate delay τ_{delay} into consideration, the accurate instantaneous depth (range) map $D(r, c)$ is physically reconstructed via the absolute ToF principle

$$D(r, c) = \frac{c}{2}\left[\hat{t}_j\left(f'\right) + \tau_{\text{delay}}\right] \quad \forall (r, c) \in \mathcal{O}_j. \tag{25}$$

Ultimately, this trajectory-driven depth reconstruction mechanism converts discrete, migrated photon events into a high-fidelity instantaneous spatial and distance representation, providing an instantaneous state for the target tracking.

E. Scale-Adaptive Multitarget Tracking Guided by Spatiotemporal Kinematics

While the preceding Stages (A–D) successfully reconstruct precise instantaneous depth and intensity representations from ultrasparse binomial triggers, dynamic UAV monitoring requires linking these macroscopic observations continuously across the focal plane. In multitarget scenarios, the relative motion between the UAVs and the SPL causes continuous variations in both transverse projection and apparent scale. To establish continuous trajectories without succumbing to localization drift, a scale-adaptive tracking mechanism is integrated directly into the MRAT framework, physically guided by the spatial ROIs and the BANSAC kinematic priors obtained in the earlier stages.

1) Initial Target Region Initialization: For each validated candidate target region $\mathcal{O}_f$ extracted from the spatiotemporal consistency verification (Stage C), a rectangular region of interest (ROI) is initialized as the tracking window

$$\mathcal{R}_i = \left[x_i, y_i, w_i, h_i\right], \quad i = 1, \ldots, n \tag{26}$$

where (x_i, y_i) denotes the ROI position and (w_i, h_i) represents its spatial extent on the Gm-APD array. This ROI provides the spatial support containing the purified target-photon returns.

2) Photon-Consistent Optical Flow Motion Estimation: Unlike conventional electro-optical images, the reconstructed frames represent localized photon-event aggregations. Assuming that the target's scattering cross section and laser illumination remain relatively stable between adjacent fast-gated frames (photon-return consistency), dense optical flow is computed within the ROI using the Farneback method to estimate the transverse motion. Let the optical flow field be denoted as

$$\mathbf{v}(x, y) = \left(v_x(x, y), v_y(x, y)\right) \tag{27}$$

which describes the pixel-wise displacement driven by the actual target projection dynamics.

The macrodisplacement of the target region is estimated by averaging the flow vectors within the filtered ROI, naturally suppressing the influence of background noise

$$(\Delta x, \Delta y) = \frac{1}{|\mathcal{R}_i|} \sum_{(x,y)\in\mathcal{R}_i} \mathbf{v}(x, y). \tag{28}$$

The transverse ROI position is updated iteratively as $(x_{t+1}, y_{t+1}) = (x_t + \Delta x,\ y_t + \Delta y)$.

3) Kinematic-Guided Failure Recovery and Scale-Adaptive Updating: As the UAV swarm performs longitudinal maneuvers (radial motion), the targets' apparent scales on the focal plane vary inversely with depth. The average magnitude of the optical flow vectors within the ROI serves as a direct indicator of this perspective scale variation

$$\bar{M}^{(t)} = \frac{1}{\left|\mathcal{R}^{(t)}\right|} \sum_{(x,y)\in\mathcal{R}^{(t)}} \left\|\mathbf{v}(x, y)\right\|. \tag{29}$$

The relative scale variation between consecutive frames is evaluated to adaptively resize the tracking window

$$\gamma^{(t)} = \sqrt{\frac{\bar{M}^{(t)}}{\bar{M}^{(t-1)} + \varepsilon_2}}, \quad \left(w^{(t)}, h^{(t)}\right) = \gamma^{(t)}\left(w^{(t-1)}, h^{(t-1)}\right) \tag{30}$$

where ε_2 is a negligible mathematical constant (e.g., 10^{-5}) included solely to prevent singularity and division-by-zero errors during periods of absolute target stasis, exerting no empirical influence on the overall tracking dynamics.

Crucially, in extreme photon regimes where signal fading or abrupt occlusion occurs, the transverse optical flow may become unstable due to insufficient photon density. Rather than relying on arbitrary search heuristics, the framework introduces a logically coupled recovery strategy. When flow estimation confidence drops, target matching is dynamically restricted to a local neighborhood strictly predicted by the BANSAC trajectory model, which preserves the statistical depth kinematics of the target, it provides a robust physical prior for the target's continuous state. By fusing the depth-kinematic prediction with the focal-plane template matching, the framework maintains stable multitarget tracking even under intermittent photon starvation, closing the processing loop from multitarget segmentation to the dynamic tracking of each individual target.

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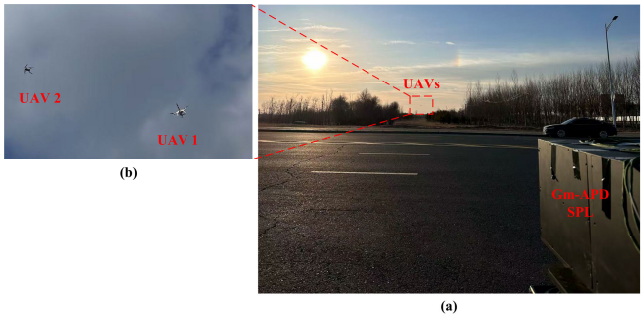


Fig. 3. Experimental SPL platform and corresponding multi-UAV field experiments. (a) SPL-based detection system deployed in the outdoor measurement scenario. (b) Dynamic multi-UAV flight experiment conducted during field testing.

IV. Field Experiment Verification of Multi-UAV Target Detection and Tracking

To comprehensively validate the effectiveness, robustness, and physical limitations of the proposed MRAT framework, the Gm-APD SPL prototype was developed and deployed to conduct field experiments at approximately 16:00 local time with ambient illuminance at the SPL entrance of about 90 530 lux and atmospheric visibility of approximately 10 km. The prototype was driven by a 1064-nm laser operating at a 20-kHz repetition rate. Photon detection was performed using a 64×64 Gm-APD focal-plane array configured with a $1^\circ \times 1^\circ$ FoV. Photon events were captured within a 2000-ns gating window at the temporal resolution of 1 ns. Up to three UAVs, including DJI Phantom 4 and Mavic Pro, were flown within the FoV up to 2 km. The UAVs performed representative maneuvers, such as hovering, trajectory crossing, longitudinal separation, and partial occlusion, evaluating multitarget sensing and tracking performance, as illustrated in Fig. 3.

The experimental campaign was systematically designed to evaluate five key capabilities of the proposed MRAT framework: 1) the limit of reliable multitarget segmentation under ultralow-SNR conditions; 2) the contribution and effectiveness of the core segmentation modules through ablation analysis; 3) radial and transverse separability limits of multi-UAV targets within the FoV under severe overlap; 4) segmentation robustness in complex meteorological and cluttered environments; and 5) dynamic multitarget tracking and kinematic performance. Together, these progressively structured experiments provide a comprehensive assessment of the segmentation and tracking reliability of MRAT in real-world photon-starved sensing scenarios.

A. Multitarget UAVs' Segmentation Capability in Ultralow-SNR Regimes

This experiment evaluates the minimum SNR at which the proposed MRAT framework can reliably segment multi-UAV targets. During data acquisition, the background photon rate remains approximately constant due to stable illumination conditions, while the signal photon rate gradually decreases as the UAVs move farther from the SPL. As a result, the effective SNR progressively degrades over sampling time.

A representative frame (Frame 3413) is selected to illustrate the segmentation process under extremely photon-starved conditions. In this frame, the average signal photons per pixel are 0.1719, while the background photons per pixel reach 3.8391, corresponding to an SNR of −13.4908 dB. After adaptive gating suppresses systematic prepulse noise, photon arrival-time histograms are constructed, and a background model is estimated using exponential fitting. The fit background distribution is then removed to obtain the residual photon-event distribution, as illustrated in Fig. 4(a). Candidate pulse peaks are first detected in the residual histogram, as the red circle illustrated in Fig. 4(b). After structural consistency verification, photon events associated with valid peaks are projected onto the detector array to identify spatially connected regions. BANSAC fitting is then applied to the photon arrival-time series within each region to estimate the ToF parameters and reconstruct both range and intensity images, as shown in Fig. 4(c) and (d).

To quantitatively evaluate segmentation robustness at low SNR, MRAT is compared with several representative reconstruction approaches, including matched filtering (MF) [32], maximum likelihood estimation (MLE) [33], peak detection filtering (PKF) [34], and kernel density estimation (KDE) [35]. All methods are applied to the same dataset with identical frame accumulation to reconstruct, as shown in Fig. 5.

As the targets move farther from the SPL, the decreasing signal photon leads to progressive degradation in reconstruction quality. PKF fails first at frame 557 (SNR −4.6040 dB), followed by MLE at frame 1447 (SNR −7.0666 dB). KDE remains functional until −10.7697 dB, while MF fails at frame 3398 when the SNR reaches −11.3427 dB. To provide a consistent evaluation criterion, the detection probability is evaluated for the smaller UAV within the FoV, which represents the more challenging target due to its weaker photon returns. The signal photon level is kept approximately constant, while the background photon count is gradually increased to emulate stronger ambient illumination, leading to progressive SNR degradation. The resulting detection probability versus SNR is shown in Fig. 6.

Notably, the proposed MRAT framework maintains reliable multitarget segmentation throughout the entire acquisition sequence even under the extreme condition of −14.0814 dB, corresponding to only 0.1744 signal photons per pixel and 4.4636 noise photons per pixel. The corresponding reconstruction results for the Phantom 4 and Mavic Pro under this ultralow-SNR scenario are presented in Fig. 7.

When the SNR decreases further, photon events triggered by the UAVs become statistically indistinguishable from background-triggered events, preventing stable convergence of the BANSAC in fitting the ToF. This regime, therefore, defines the practical segmentation boundary of the MRAT framework under the current SPL configuration.

B. Ablation Analysis of the Core MRAT Architecture

A systematic ablation study is conducted on the same representative Frame 3413 to analyze the individual contributions of the core front-end modules in MRAT, as shown in Fig. 8. Since segmentation quality directly determines the reliability of the downstream tracking stage (Stage V), the first four processing stages are individually removed to assess

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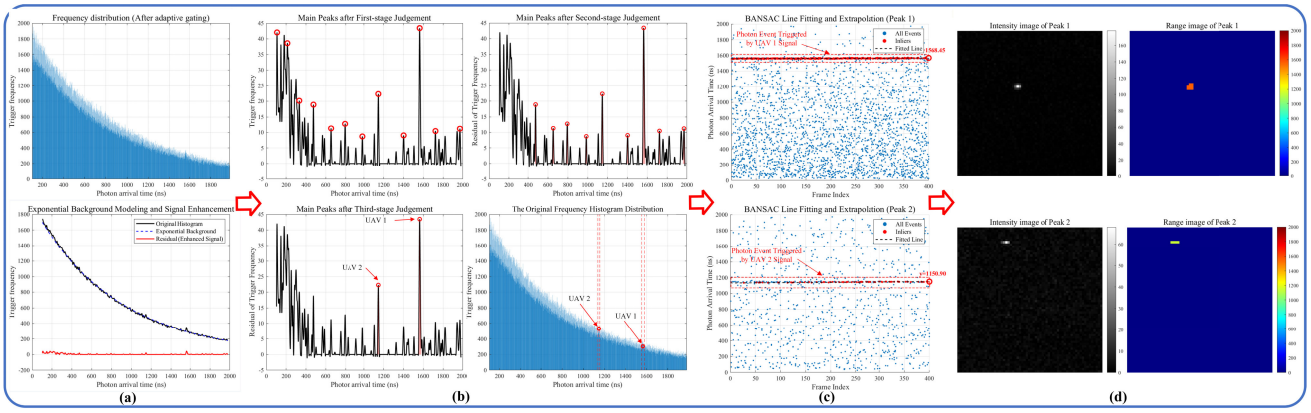


Fig. 4. MRAT-based target segmentation procedure: (a) Global histogram of photon arrival times and background-noise modeling, including the fit background-triggered photon-event distribution and the residual distribution. (b) Third-order residual peak detection to determine the photon-arrival-time range associated with target-triggered events and to extract candidate targets. (c) Photon-arrival-time fitting and distance reconstruction for the candidate targets. (d) Intensity and range reconstruction results of the multi-UAV targets.

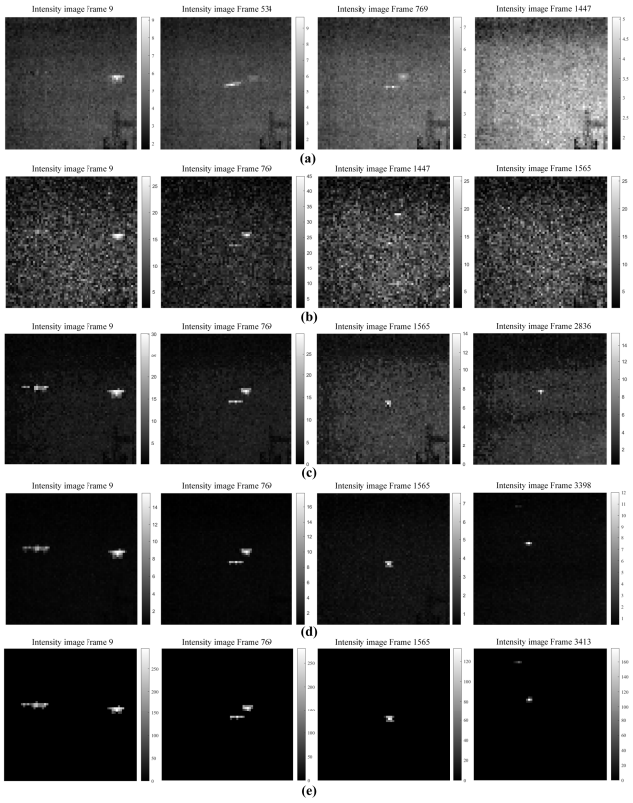


Fig. 5. Multiframe intensity reconstructions under progressively decreasing SNR conditions using different methods. The failure boundary for each method is observed at (a) −4.6040 dB for MLE, (b) −7.0666 dB for PKF, (c) −10.7697 dB for KDE, (d) −11.3427 dB for MF, and (e) −14.0814 dB for the proposed MRAT framework.

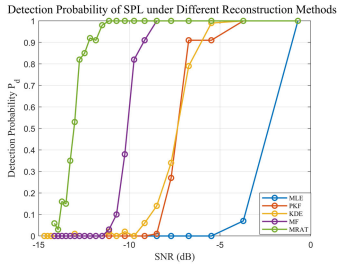


Fig. 6. Detection probability for multitarget detection and segmentation using different methods.

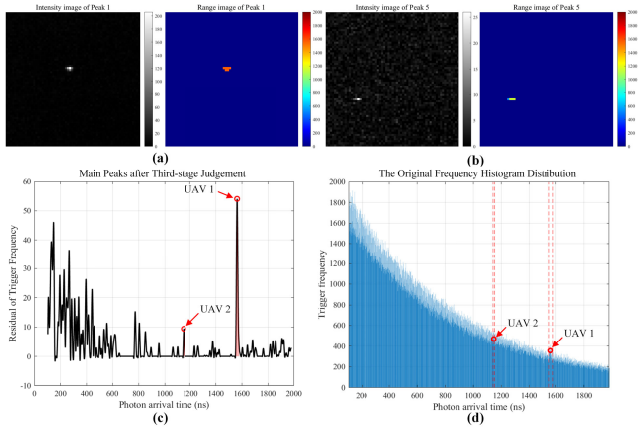


Fig. 7. Multitarget segmentation and reconstruction results at the practical operating limit of the proposed MRAT framework (−14.0826 dB). (a) and (b) Segmentation and reconstruction results for the Phantom 4 and Mavic Pro UAVs. (c) Third-order residual peak detection results. (d) Corresponding temporal positions are distributed in the trigger frequency histogram.

their respective impacts on target extraction and reconstruction performance. The baseline result obtained using the complete MRAT pipeline is presented in Fig. 8(a).

For Stage I, disabling adaptive suppression fundamentally distorts the temporal statistical baseline, as shown in Fig. 8(b). The retained hardware-induced prepulse transient significantly expands the histogram dynamic range, thereby compressing the relative signal-to-background contrast of weak UAV echoes. Moreover, the steep decay tail introduces strong local gradients, under which stochastic Poisson fluctuations are erroneously amplified into dense false detections. Although the far-field targets in this frame remain partially observable due to their separation from the transient-dominated region, targets located within high-gradient intervals would be completely obscured. This confirms that Stage I is indispensable for baseline stabilization and gradient-induced false-alarm suppression.

Stage II directly determines target detectability. As shown in Fig. 8(c), removing residual-enhanced filtering causes a

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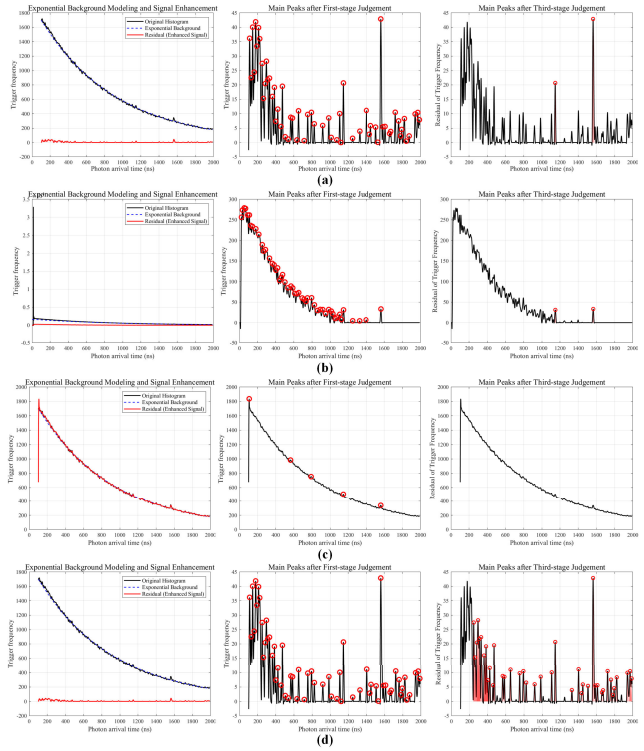


Fig. 8. Temporal ablation analysis of the MRAT framework under photon-starved conditions. (a) Full pipeline. (b) Removal of Stage I causes the detection of more candidate pulse peaks. (c) Removal of Stage II leads to target suppression by the noise floor. (d) Removal of Stage III produces dense spurious peaks.

complete collapse of target extraction. Without explicit modeling and subtraction of the nonstationary background, peak detection operates directly on the raw photon-triggering statistics, where stochastic Poisson fluctuations dominate the temporal distribution. Consequently, random fluctuations are falsely identified as candidate peaks, while the true target echoes (e.g., around ∼1200 and ∼1550 ns) remain entirely submerged beneath the noise floor. Under these conditions, no pulse peaks consistent with the expected UAV signal characteristics can be reliably recovered.

While Stage II ensures temporal separability, Stage III enforces spatial physical validity. As shown in Fig. 8(d), removing the spatiotemporal consistency verification eliminates the CFAR and connected-component constraints required for rejecting spatially incoherent events. As a result, randomly distributed dark counts and background photons are incorrectly retained as valid pulse detections, producing a dense proliferation of spurious temporal peaks. When projected into the spatial domain, these false detections reconstruct multiple nonphysical targets, as illustrated in Fig. 9. This demonstrates that Stage III is essential for preserving spatial reconstruction fidelity and maintaining a low false-alarm rate.

When Stage IV is ablated, the system defaults to a conventional long-integration TCSPC strategy, in which all triggered photon events are aggregated over an extended temporal window to estimate depth. For high-dynamic UAVs, such prolonged integration inevitably introduces severe motion blur along the range dimension. More critically, this baseline

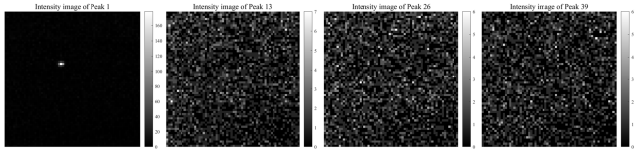


Fig. 9. Representative intensity reconstruction artifacts generated from random noise-induced pulse peaks after deactivating the spatiotemporal consistency verification module, illustrating the proliferation of nonphysical target structures caused by uncoordinated photon-trigger events.

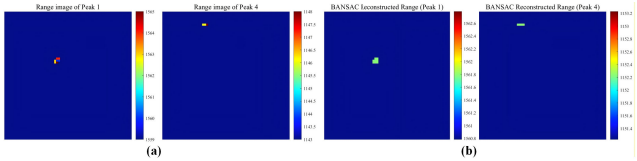


Fig. 10. Ablation analysis of depth reconstruction methods. (a) Conventional long-integration TCSPC without BANSAC, resulting in the temporally averaged historical position. (b) Proposed MRAT with Stage IV, accurately reconstructing the instantaneous state.

approach yields only a temporally averaged position, resulting in inherent latency. In contrast, the proposed BANSAC module explicitly fits the spatiotemporal trajectories of photon events and extrapolates the depth to the end of the sampling frame. As shown in Fig. 10, this kinematic reconstruction not only eliminates intratarget depth variation—producing a compact and coherent spatial footprint—but also accurately recovers the instantaneous target position at the end of the acquisition window. Such a zero-lag kinematic state is essential for reliable initialization and update of downstream tracking.

Overall, the ablation results confirm that all stages are indispensable under photon-starved and low-SNR conditions. Among them, residual-enhanced filtering (Stage II) fundamentally determines the detection limit, as it directly governs whether target photons can be separated from the noise floor. Without this stage, no subsequent processing can recover the targets. In contrast, adaptive suppression (Stage I) does not directly enhance detectability but ensures statistical stability. The hardware-induced prepulse transient distorts the global distribution, inflating the mean and variance and degrading both background modeling and CFAR thresholding. Without suppression, this leads to severe false alarms and threshold failure, especially in high-background or near-field conditions. Therefore, Stage II determines sensitivity, while Stage I guarantees dynamic range in practical scenarios.

C. Radial and Transverse Separability Limits of Multi-UAV Targets Within the FoV Under Severe Overlap

Reliable multi-UAV perception under ultrasparse photon conditions fundamentally depends on target separability within the FoV. In practical scenarios, separability is jointly constrained by projection overlap in the transverse spatial domain and temporal overlap in the radial photon-arrival-time domain. To investigate these limits, a series of experiments was conducted to evaluate the transverse and radial separability boundaries of the proposed MRAT framework under severe multitarget overlap conditions.

First, transverse segmentation capability was evaluated in scenarios where multiple UAVs overlap in the projection plane

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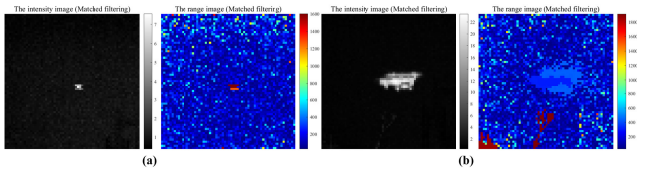


Fig. 11. Reconstruction results under multiple-target projection overlap for (a) far- and (b) near-field scenarios.

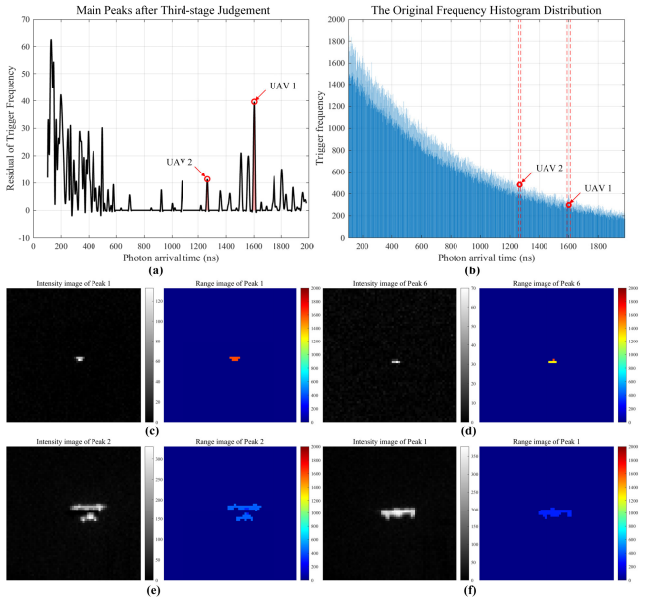


Fig. 12. MRAT-based segmentation and reconstruction results under severe UAV projection overlap. (a) and (b) Temporal residual-peak fitting results and the corresponding triggering photon-event frequency position distributions for far-field UAV targets. (c) and (d) Processing results for far-field targets. (e) and (f) Processing results for near-field targets.

while remaining separated along the range dimension. Such configurations are particularly challenging for conventional image-centric SPL reconstruction methods because spatial overlap produces merged or blurred intensity responses. As shown in Fig. 11, existing reconstruction methods fail to generate spatially separable targets under both far- and near-field conditions, making it difficult to reliably determine the number of UAVs within the FoV.

By exploiting third-order residual peak detection, MRAT identifies distinct temporal pulse positions associated with different targets, thereby enabling reliable segmentation even when UAVs are partially or fully occluded in the transverse plane, as illustrated in Fig. 12(a) and (b). In the far-field scenario (0.6013 signal photon per pixel, 4.1082 noise photon per pixel, and SNR ≈ -8.35 dB), MRAT successfully separates the overlapping targets. The reconstructed range and intensity images shown in Fig. 12(c) and (d) yield estimated distances of 312.36 and 264.00 m for the Phantom 4 and Mavic Pro UAVs, respectively.

A near-field experiment with 1.8596 signal photons per pixel, 4.0673 noise photons per pixel, and SNR ≈ -3.40 dB further confirms the robustness of MRAT. As shown in Fig. 12(e) and (f), the proposed framework consistently resolves partially occluded UAVs and reconstructs their range and intensity distributions with high fidelity. These results demonstrate that MRAT maintains reliable transverse segmentation capability even under severe projection overlap and low-SNR conditions.

Beyond transverse overlap, the intrinsic radial resolution limit of the proposed MRAT framework was further investigated using three UAVs progressively approaching each other along the depth dimension. When the longitudinal separation between adjacent UAVs remained larger than approximately 4.5 m, corresponding to about 30 ns in photon arrival-time difference [36], MRAT reliably detected and reconstructed all targets. Distinct residual peaks were observed, and BANSAC fitting of the triggered photon-event temporal distributions produced stable ToF estimates, as illustrated in Fig. 13(a) and (b). It is worth noting that the adaptive suppression stage removes approximately 170 ns of hardware-induced prepeak interference from the effective temporal observation window. Under the current 2000-ns SPL gating interval, the remaining usable temporal span, therefore, theoretically supports the separation of up to approximately 61 UAV targets along the FoV direction based on the experimentally observed 30-ns minimum resolvable photon-arrival-time interval.

As the longitudinal separation decreased below this threshold, photon events originating from different UAVs increasingly overlapped in the temporal domain. Under these conditions, the residual peaks gradually merged, causing the BANSAC fitting process to fail in resolving individual targets. Consequently, ambiguous intensity reconstructions and segmentation artifacts appeared, as shown in Fig. 13(c) and (e). This phenomenon defines the intrinsic temporal-resolution limit under severe radial overlap conditions.

It is worth noting that the range-gating mechanism of SPL enables the photon-event statistics to be selectively accumulated within a predefined depth interval, thereby effectively suppressing a large fraction of background-triggered photon events. However, when static echoes such as tree branches fall within the same gating window as the UAVs, they produce temporally stable photon returns that cannot be distinguished from UAV echoes through residual-peak analysis alone. In our experiments, tree branches located at altitudes of approximately 10 m generated stable and well-defined pulse peaks in the photon events frequency histogram, as shown in Fig. 14. Under this condition, the third-order residual detector in Stage II incorrectly interpreted the static echoes as additional UAV targets, leading MRAT to erroneously identify and reconstruct false targets.

D. Segmentation Robustness in Complex Meteorological and Cluttered Environments

To further assess the performance of the proposed MRAT under adverse weather conditions and complex background environments, we designed three representative experimental scenarios: 1) a rainy scene with a forest background (visibility: 1.5 km and relative humidity: 20%–55%); 2) a foggy scene with an urban-building background (visibility: 2.5 km and relative humidity: 10%–75%); and 3) a foggy scene with an open-sky background (visibility: 2.5 km and relative humidity: 10%–75%).

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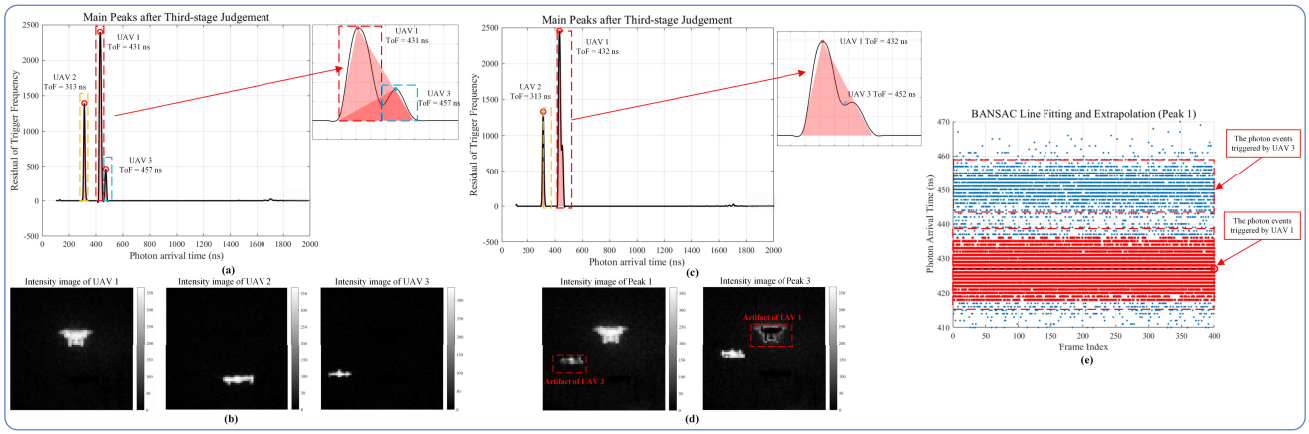


Fig. 13. Time-domain fitting and intensity reconstruction results processed by the MRAT framework for three near-field UAV targets. (a) Third-order residual-peak fitting results for determining the temporal frequency-position distributions of three overlapping UAVs. (b) Intensity reconstruction results obtained using the proposed MRAT framework. (c) Failure case of third-order residual-peak fitting caused by severe peak overlap. (d) BANSAC fitting failure under insufficient radial separation. (e) Intensity reconstruction artifacts caused by unresolved temporal overlap.

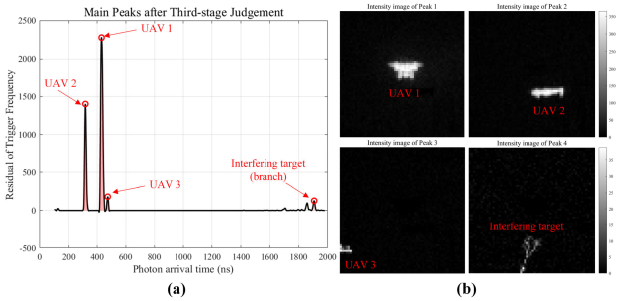


Fig. 14. Failure cases of MRAT segmentation under persistent static interference. (a) Degradation of third-order residual-peak segmentation caused by interfering static echoes. (b) Erroneous segmentation results caused by interference-induced stable photon returns, leading to four falsely identified targets.

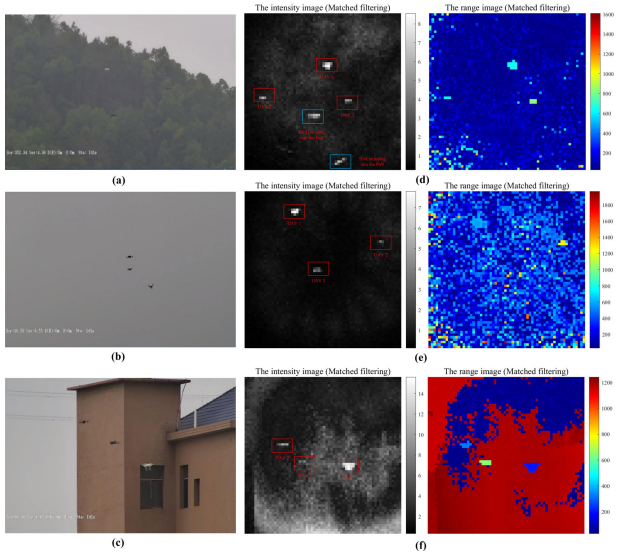


Fig. 15. Electro-optical imaging and MF-based SPL reconstructions under three weather conditions: (a)–(c) electro-optical images and (d)–(f) corresponding SPL reconstructions for rainy forest, foggy urban-building, and foggy open-sky scenes.

Three UAVs (DJI Matrice 4E, Phantom 4, and Air 2) performed maneuvering flights within a maximum slant range of approximately 500 m at an altitude of about 50 m, while a synchronized electro-optical imaging system recorded reference images, as shown in Fig. 15(a)–(c).

For reference, Fig. 15(d)–(f) presents the 400-frame accumulated reconstructions obtained using a conventional matched filtering pipeline. Under adverse weather conditions such as fog and rain, scattering from water vapor and raindrops significantly increases the probability of background-noise-triggered photon events occurring prior to the temporal gating window. As a result, both the intensity and depth reconstructions exhibit elevated noise levels and dense speckle artifacts. Even at relatively short ranges, the imaging quality deteriorates substantially, making it difficult to reliably determine the number of UAVs present within the FoV. This limitation becomes more pronounced in scenarios involving additional moving objects. For example, when birds accompany the UAVs [see the blue-marked region in Fig. 15(d)], MF-based reconstructions suffer from severe ambiguity, further degrading target discrimination and often leading to false or merged detections.

Despite these challenges, MRAT successfully segments all three UAVs by exploiting the 4-D photon-event representation and residual peak analysis. The segmentation results for the three scenarios are shown in Fig. 16.

The fit photon arrival-time distributions of MRAT for each UAV are illustrated in Fig. 17. Fast-moving birds do not produce stable temporal peaks due to their rapid motion relative to the SPL sampling rate and are, therefore, suppressed as noise during residual peak detection.

However, when buildings and UAVs fall within the same gating window, backscattered photons from the buildings form a spatially extensive and temporally stable connected component in the photon-event domain. The resulting persistent residual peaks cannot be suppressed by third-order analysis, as shown in Fig. 18. This phenomenon mirrors the "tree-branch interference" case: background reflectors with stable structure and temporal consistency generate enduring return peaks that cannot be removed by residual-peak detection, preventing reliable operation of MRAT in such scenarios.

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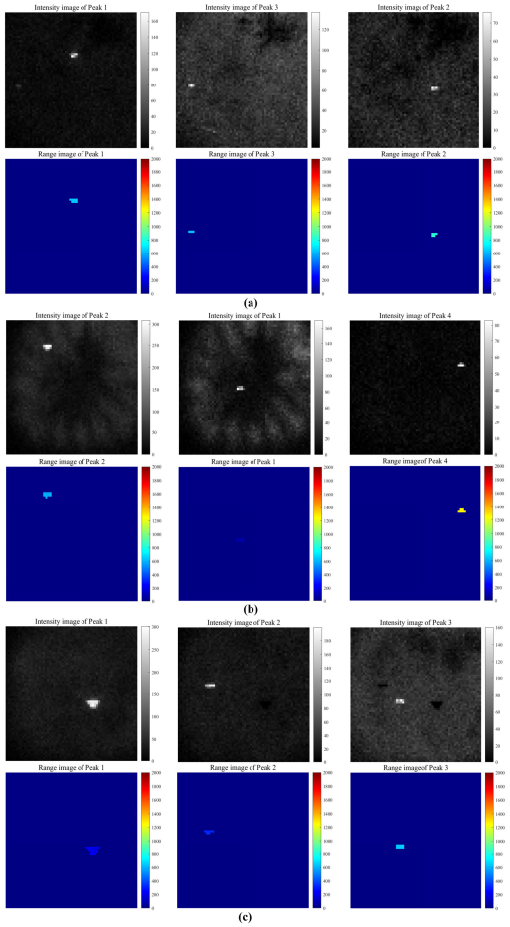


Fig. 16. Segmentation results of near-field UAVs under different weather conditions using the proposed MRAT algorithm: (a) rainy forest, (b) foggy urban building, and (c) foggy open sky.

E. Multitarget Dynamic Tracking and Kinematic Evaluation

The dynamic tracking capability of the proposed MRAT framework is further evaluated based on the designed scale-adaptive tracking module, hereafter referred to as OpticalFlow. Five representative tracking algorithms are selected for comparison: particle filter (PF) [37], mean-shift Kalman fusion (mean-Kalm) [38], joint probabilistic data association (JPDA) [39], multiple hypothesis tracking (MHT) [40], and probability hypothesis density (PHD) [41].

To ensure a strictly fair and rigorous evaluation, all tracking methods are driven by the same MRAT-derived intensity reconstructions. The unified min–max intensity normalization is applied, whereby photon-triggering frequencies associated with valid signal pixels within each detected pulse are independently scaled to the intensity region $[0, 1]$. The only distinction lies in whether target-level segmentation is explicitly preserved or recombined, in accordance with the intrinsic capabilities of each tracker. For trackers without inherent multitarget capability (i.e., MRAT tracking module, PF, and mean-Kalm), tracking is performed directly on these segmented, normalized single-target intensity maps. In contrast, for multitarget tracking algorithms (i.e., JPDA, MHT, and PHD), the same normalized target-level reconstructions are first obtained via MRAT segmentation and then recombined into a single frame by pixelwise accumulation across all targets. Through this "segment-normalize-reintegrate" strategy, the intensity distribution and geometric features of every target within the composite frame remain perfectly identical to those in the isolated segmented inputs. Consequently, any observed performance differences arise solely from the data association and motion modeling mechanisms of the trackers, rather than from discrepancies in input data representation.

Tracking experiments were conducted in both near- and far-field multi-UAV scenarios to comprehensively evaluate the robustness of the proposed framework under different photon conditions. Tracking accuracy was quantified using the Euclidean distance between the tracking trajectory and the manually annotated position in each frame. The evaluation metrics include the mean tracking error (Mean), standard deviation (STD), and root mean square error (RMSE)

$$\begin{cases} e_t = \sqrt{(\hat{x}_t - x_t)^2 + (\hat{y}_t - y_t)^2} \\ \text{Mean} = \dfrac{1}{T}\displaystyle\sum_{t=1}^{T} e_t \\ \text{STD} = \sqrt{\dfrac{1}{T}\displaystyle\sum_{t=1}^{T} (e_t - \text{Mean})^2} \\ \text{RMSE} = \sqrt{\dfrac{1}{T}\displaystyle\sum_{t=1}^{T} e_t^2} \end{cases} \tag{31}$$

where $(\hat{x}_t, \hat{y}_t)$ denotes the estimated target center in the tth frame and (x_t, y_t) represents the corresponding truth position.

In the near-field scenario, three UAVs (one DJI Phantom 4, one DJI Mavic Pro, and one DJI Mini) performed dynamic flight maneuvers at ranges of up to 200 m. The corresponding MRAT-based segmentation and tracking results are shown in Fig. 19. Although reliable three-target tracking is achieved, the relatively strong photon returns under near-field conditions do not fully reflect the detection and tracking limit of the SPL.

To further evaluate tracking robustness under extreme photon-starved conditions, a more challenging far-field experiment was conducted at a range of 500 m. Due to severe signal attenuation with increasing distance, two UAVs (one DJI Phantom 4 and one DJI Mavic Pro) remained distinguishable within the FoV; however, this scenario approaches the practical operating limit of the SPL system and, therefore, serves as the primary benchmark for evaluating tracking performance under ultralow-SNR conditions. Tracking remains stable until the MRAT framework reaches its segmentation limit of Frame 3413 at an extreme condition of −14.0814 dB, corresponding to 0.1744 signal photons per pixel and 4.4636 noise photons per pixel. The segmentation and tracking results under this ultralow-SNR scenario are presented in Fig. 20.

Quantitative tracking error statistics for the far-field scenario are summarized in Table I. For Target 1, OpticalFlow achieves a mean error of 0.9299 pixels and an RMSE of 1.1396 pixels, corresponding to reductions of approximately 97.3%/97.6% and 20.5%/67.2% in Mean/RMSE compared with PF and mean-Kalm, respectively. For Target 2, OpticalFlow maintains subpixel tracking accuracy with a mean error of 0.6566 pixels and an RMSE of 0.7948 pixels, yielding improvements of approximately 26.9%/44.1% relative to PF and 43.9%/74.9%

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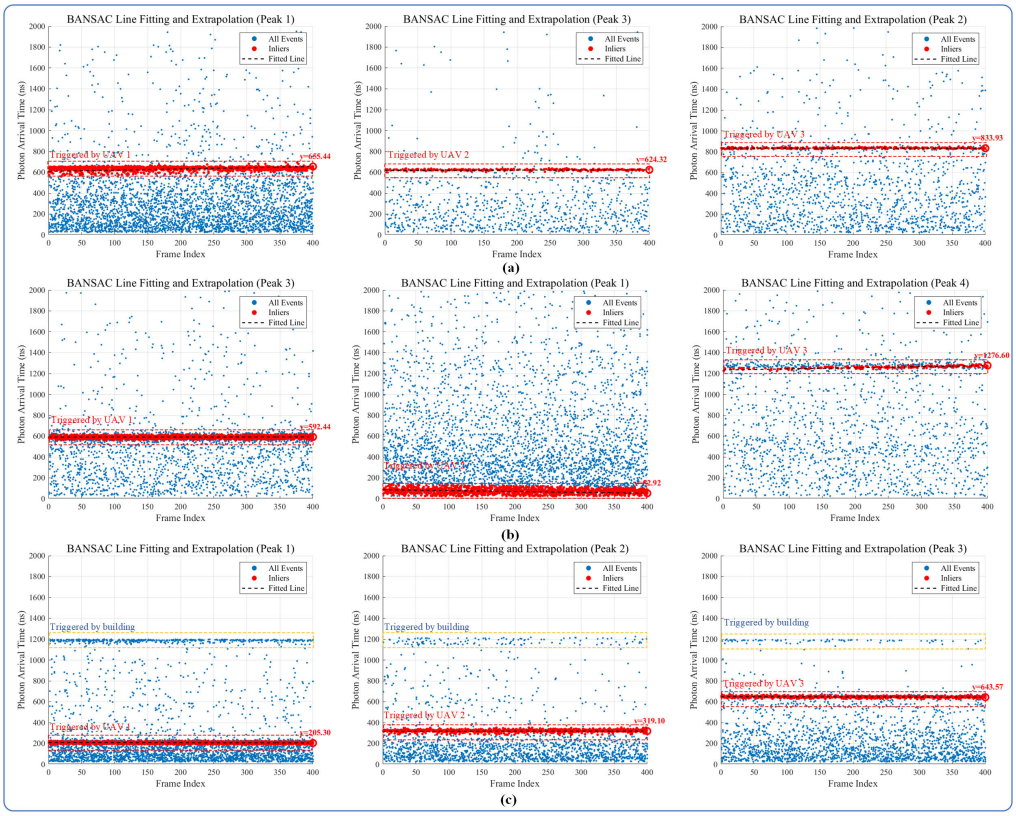


Fig. 17. Fit photon arrival-time distributions of the segmentation results of near-field UAVs under different weather and background conditions using the proposed MRAT algorithm. (a) Rainy forest scenario. (b) Foggy open-sky scenario. (c) Foggy urban-building scenario.

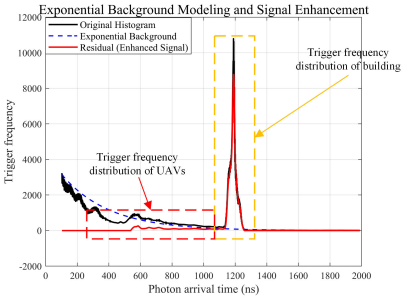


Fig. 18. Segmentation failure due to stable background interference when both UAVs and urban buildings fall within the same gating window, illustrating persistent residual peaks in the photon-event domain.

TABLE I

TRACKING ERRORS COMPARISON FOR FAR-FIELD MULTI-UAV USING DIFFERENT TRACKING ALGORITHMS

UAV	Method	Mean	STD	RMSE
UAV 1	OpticalFlow	0.9299	0.6604	1.1396
	PF	34.8196	32.6187	47.6562
	Mean-Kalm	1.1703	3.2809	3.4757
	JPDA	1.1896	4.6156	4.7553
	MHT	4.7535	9.5115	10.6121
	PHD	2.2638	6.8492	7.1975
UAV 2	OpticalFlow	0.6566	0.4490	0.7948
	PF	0.8979	1.1051	1.4218
	Mean-Kalm	1.1695	2.9536	3.1699
	JPDA	2.3059	9.8774	10.1191
	MHT	3.1195	8.9425	9.4501
	PHD	15.7063	18.7086	24.3919

relative to mean-Kalm in terms of Mean/RMSE, respectively. Although JPDA, MHT, and PHD possess inherent

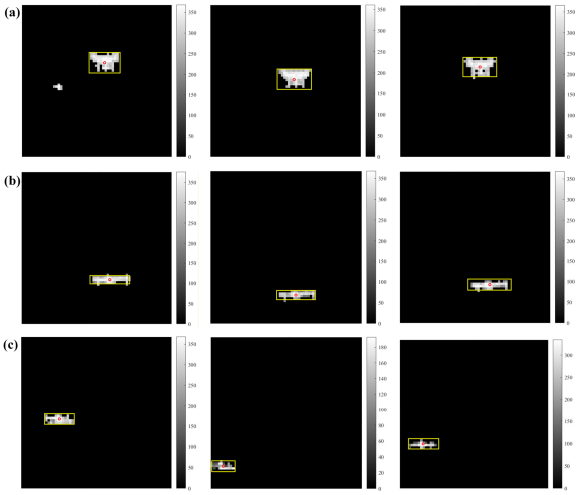


Fig. 19. MRAT-based segmentation and tracking results for near-field multi-UAV targets. (a) DJI Phantom 4, (b) DJI Mavic Pro, and (c) DJI Mini.

multitarget tracking capability, all three methods suffer from frequent target loss and fragmented trajectory association under photon-starved conditions, leading to substantial performance degradation. It should be noted that, for JPDA, MHT, and PHD, the longest continuously associated trajectory segment is selected for quantitative error evaluation to ensure a fair comparison.

Meanwhile, the robustness of the proposed MRAT tracking framework is further evaluated by analyzing the tracking error of UAV 1 under progressively degraded SNR conditions.

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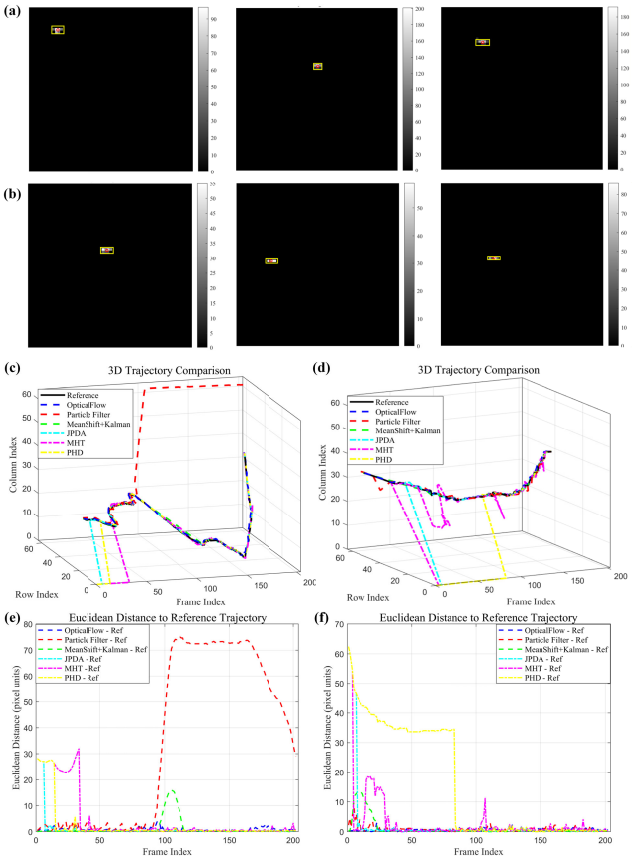


Fig. 20. MRAT-based tracking results for two far-field UAV targets as obtained from different tracking algorithms. (a) and (b) MRAT-based segmentation and intensity reconstruction of far-field UAV targets. (c) and (d) Dynamic tracking trajectories. (e) and (f) Corresponding tracking error distributions.

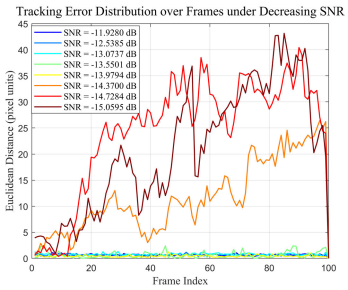


Fig. 21. Tracking error variation of UAV 1 as a function of SNR under a fixed signal photon level.

During this experiment, the average signal photons per pixel are maintained approximately constant, while the background photon level is gradually increased. As shown in Fig. 21, both the mean tracking error and its variance increase monotonically as the SNR decreases, indicating growing uncertainty in target tracking and localization under stronger background noise. When the SNR approaches approximately −14 dB under the given signal-photon level, the tracking error exhibits pronounced fluctuations, and trajectory drift and intermittent mistracking frames begin to emerge.

In addition to tracking accuracy, the real-time capability and scalability of the proposed MRAT framework are also evaluated. For the employed 64 × 64 Gm-APD array, each frame contains 16 384 bytes of raw photon-count data, corresponding to a throughput of approximately 409.6 MB/s at the maximum acquisition rate of 25 kHz. Considering both data-transfer and computational overhead, the complete MRAT pipeline achieves a stable end-to-end visualization rate of approximately 50 Hz under a 20-kHz laser repetition frequency, thereby satisfying the real-time requirements of dynamic UAV trajectory monitoring.

The current effective detection range of the SPL is limited to approximately 1 km, primarily due to the relatively large receiving FoV (1° × 1°), which spatially reduces the photon density return on the detector plane. Based on the LiDAR link equation and the Gm-APD triggering probability model, reducing the FoV to the sub-0.1° regime would significantly increase the received photon density at long distances while preserving the same photon accumulation strategy. Under such conditions, the proposed MRAT framework is expected to maintain comparable segmentation and tracking performance at ranges approaching 15 km, even for complex multi-UAV swarm scenarios. This indicates that the proposed framework possesses strong scalability and substantial potential for long-range photon-efficient UAV monitoring applications.

V. Conclusion

We propose an MRAT framework for UAV swarm detection and tracking using SPL under ultralow SNR and severe overlap of target spatial positions. Field experiments demonstrate that MART achieves reliable multitarget segmentation and tracking at SNRs below −14 dB, with an average of less than 0.2 photons per pixel. The method resolves targets with a minimum radial separation of 4.50 m and maintains tracking errors below 0.92 pixels. Compared with mean-Kalm and particle filter baselines, MART reduces tracking errors by more than 26% and 43%, respectively, demonstrating strong robustness in both near- and far-field scenarios.

By exploiting the statistical structure of photon-arrival events, MRAT effectively addresses the long-standing challenge of multitarget perception under ultrasparse and temporally overlapped photon-return conditions. Unlike conventional image-centric approaches, the proposed framework operates directly on temporal photon-event statistics, enabling reliable segmentation and tracking even when reconstructed intensity features become unreliable. Since the MRAT framework is fundamentally independent of specific SPL configurations, its sensing capability can be further extended through optical-system optimization without modifying the underlying algorithmic architecture. Link-budget analysis suggests that reducing the optical FoV to the sub-0.1° level could substantially improve long-range photon efficiency, potentially extending reliable multitarget perception to ten-kilometer-scale sensing ranges. These results demonstrate the potential of photon-statistics-driven processing for scalable UAV swarm monitoring and low-altitude situational awareness under extreme photon-starved conditions.

Despite these advantages, the current MRAT framework still exhibits limitations in scenarios where static background objects generate persistent photon returns within the same gating window as multi-UAV targets. Future work will focus on

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incorporating additional discriminative cues, including motion-induced depth variations and target scattering characteristics, to further improve robustness in cluttered environments and extend the applicability of the proposed framework to more complex operational scenarios.

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